

A cross-chain bridge between Ethereum and Acki Nacki, secured end-to-end by zero-knowledge proofs.

i Note

This guide describes the current **test deployment** (Ethereum Sepolia testnet). Features and addresses may change. Always confirm the active contract address with the team before bridging anything you care about.

1. What is the Acki Nacki Bridge?

The Acki Nacki Bridge lets you move value between the **Ethereum** network and the **Acki Nacki** blockchain. Unlike traditional bridges that rely on a trusted group of signers, this bridge uses **zero-knowledge (ZK) cryptographic proofs** to verify that every cross-chain event really happened.

You don't have to trust an operator. The bridge proves, mathematically, that your deposit was recorded on Ethereum before any tokens are issued on Acki Nacki.

Two directions, two mechanisms

Direction	What it does	Status for users
Ethereum → Acki Nacki (Deposit)	You send ETH on Ethereum; equivalent tokens are minted to you on Acki Nacki after a ZK proof of your deposit is verified.	✅ Available
Acki Nacki → Ethereum (Withdraw / burn)	Burn tokens on Acki Nacki and release ETH on Ethereum.	⚠️ In development — see §6

⚠️ Important

Only the Deposit direction works today. Genuine cross-chain withdrawals are still being built. Any “Withdraw” control in the current web app is a **non-functional placeholder** — even if it shows a success message, it does **not** move funds back to Ethereum.

2. Before you start

You will need:

1. **A MetaMask wallet** (browser extension or mobile). Other EIP-1193 wallets may work, but MetaMask is the supported and tested option.
2. **Test ETH on the Sepolia network.** The bridge currently runs on Ethereum’s **Sepolia testnet**, not on mainnet. Sepolia ETH has no real monetary value.
3. A few minutes of patience — proof generation and verification happen automatically in the background but are not instant.

Network details (testnet)

Parameter	Value
Network name	Sepolia
Chain ID	11155111 (0xaa36a7)
Public RPC	https://rpc.sepolia.org
Block explorer	https://sepolia.etherscan.io
Bridge contract	0xDE8180911Ab2EbC9A6c1F5526bCE4c8242C061d9

Getting Sepolia test ETH

Obtain free Sepolia ETH from public faucets (search “Sepolia faucet”). You only need a small amount to try the bridge — for example, 0.01–0.1 ETH.

3. Depositing ETH (Ethereum → Acki Nacki)

This is the main user flow.

What you do

1. **Open the Bridge web app** in a browser with MetaMask installed.
2. **Connect your wallet.** Click [Connect Wallet](#) in the header and approve the MetaMask prompt.
3. **Switch to Sepolia.** If you're on another network, the app asks MetaMask to switch automatically. Approve it.
4. **Enter the amount** of ETH to bridge in the Deposit form.
 - The amount must be **greater than 0** and **at most 100 ETH** (`MAX_DEPOSIT_AMOUNT`). Deposits of 0 or over 100 ETH are rejected on-chain.
5. **Submit and confirm** the transaction in MetaMask. This calls the bridge's `deposit()` function and sends your ETH to the bridge contract.
6. **Wait for the transaction to be mined.** Once confirmed, the bridge emits a `Deposit` event containing your unique **Deposit ID**, your address, the amount, and a timestamp.

What happens behind the scenes

Fully automated — you do none of this yourself:

1. An **off-chain prover** reads your deposit transaction and its inclusion proof from Ethereum, then builds a **Halo2 ZK proof** that your `Deposit` event was genuinely emitted by the bridge contract in a real Ethereum block.
2. The Acki Nacki side **verifies that proof natively** (via a dedicated TVM verification opcode) and checks the public inputs.

3. Once the proof is accepted, the corresponding **tokens are minted to you on Acki Nacki**.

The result: your ETH is locked on Ethereum, and you receive matching tokens on Acki Nacki — without trusting any human operator.

Tracking your deposit

- Use your **transaction hash** on the Sepolia explorer to confirm the deposit landed.
 - Note your **Deposit ID** from the `Deposit` event — it uniquely identifies your bridging operation.
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4. Checking bridge status & history

The web app surfaces live, read-only information from the contract:

- **Total deposits** made through the bridge (`depositCount`).
- **Total value deposited** (`totalDeposited`).
- **Your transaction history**, with direct links to Etherscan.

These are informational and never require a signature.

i Note

In the current build, parts of the transaction-history panel may show **sample/mock entries** (including a “Withdraw” row) for layout purposes. Trust the block explorer as the source of truth for your real transactions.

5. Fees, limits and safety

- **Per-deposit limit:** greater than 0 and up to 100 ETH per `deposit()` call.
- **Gas:** You pay normal Ethereum (Sepolia) gas fees for the deposit transaction.
- **Pausing:** Deposits are blocked while the contract is paused by the owner (`whenNotPaused`).
- **Treasury & yield:** Idle ETH held by the bridge can optionally be put to work in AAVE V3 for yield by the bridge owner. This is an

operator-side feature and does not affect your deposit balance or your claim to bridged tokens.

- **Non-custodial trust model:** Cross-chain state advances only when valid ZK proofs are verified on-chain — no trusted signer set.

Safety checklist

-  Confirm you are on **Sepolia** (testnet) and using the **correct contract address**.
 -  Never enter your seed phrase anywhere — MetaMask never asks for it in a dApp.
 -  Double-check the amount before confirming in MetaMask.
 -  This is a **testnet deployment**. Do not bridge mainnet ETH or treat test tokens as having real value.
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6. Withdrawals (Acki Nacki → Ethereum) — current status

True cross-chain withdrawals (burning tokens on Acki Nacki to release ETH on Ethereum) are **not yet available**. The legacy refund-style **withdraw** flow from earlier versions has been **retired**, and the production burn-proof flow is still under active development.

What this means right now:

- You **can** deposit ETH and receive tokens on Acki Nacki.
- You **cannot** yet move value back from Acki Nacki to Ethereum through the bridge.
- The “Withdraw” form in the current app is a **simulation only**: it animates a fake “proof generation” and may display “Withdrawal Successful!”, but **no funds are moved**. Do not rely on it.

This guide will be updated when withdrawals go live.

7. How the bridge stays trustworthy (plain-English overview)

- **Your deposit is proven, not asserted.** A ZK proof shows your deposit transaction was included in a real Ethereum block before any tokens are minted.
- **Acki Nacki's own state is attested with proofs too.** The reverse channel checks cryptographic attestations of Acki Nacki blocks (validator signatures, block identity, chain progression) using ZK proofs verified on Ethereum via `verifyBlock(...)`.
- **Strict on-chain rules.** Every state update enforces invariants such as monotonic block sequence numbers and chain-anchor consistency, so the bridge can't be tricked into accepting out-of-order or forged state.

You don't need to understand the cryptography to use the bridge — but it's the reason the bridge can be trusted by math rather than by people.

8. Troubleshooting

Problem	Likely cause	What to do
"Please connect your wallet first"	MetaMask not connected	Click Connect Wallet and approve.
"Please switch to Sepolia network"	Wrong network selected	Approve the network switch in MetaMask, or add Sepolia manually.
"Please enter an amount" / "Invalid amount"	Empty or non-numeric amount	Enter a positive number of ETH (≤ 100).
Transaction fails / reverts	Amount is 0 or over 100 ETH, insufficient gas, or paused contract	Check your Sepolia ETH balance, use $0 < \text{amount} \leq 100$, and retry.
Deposit confirmed but no tokens on Acki Nacki yet	Proof generation/verification still in progress	Wait — minting happens after the ZK proof is verified.
MetaMask not detected	Extension missing or disabled	Install/enable MetaMask and reload the page.
"Withdraw" appears to succeed but no ETH arrives	Withdraw is a non-functional placeholder	Ignore it — withdrawals are not live yet (\$6).

9. Glossary

- **ZK proof (zero-knowledge proof):** A cryptographic proof that a statement is true without revealing extra information. Here, it proves your deposit really happened.
- **Halo2 / Groth16:** ZK proof systems used by the bridge. Halo2 proofs are sometimes wrapped in compact Groth16 proofs to fit Ethereum's contract size limits.
- **Deposit ID:** A unique number assigned to each deposit, emitted in the `Deposit` event.
- **Sepolia:** An Ethereum test network used for the current deployment.
- **TVM:** The virtual machine Acki Nacki uses to execute contracts, including native proof verification.
- **Treasury:** ETH held by the bridge contract from user deposits.
- **AAVE V3:** A lending protocol where idle treasury ETH may be placed to earn yield (operator-managed; does not affect your funds' claimability).

This document describes the current test deployment of the Acki Nacki Bridge and will evolve as features such as withdrawals become available. Always verify network and contract details with the team before bridging.